

# ScreenSense: Kids' Screen Time Visualization

## 1. Project Overview

ScreenSense is a data visualization project designed to analyze digital screentime patterns among children and young adults. With digital devices becoming a central part of entertainment, learning, and communication, understanding screentime behaviour has become increasingly important. This project explores how digital device usage influences daily routines, health, and productivity, and provides insights that support the development of healthier digital habits.

This project uses an interactive Power BI dashboard to explore usage patterns across age, gender, and location, identify health impacts, compare educational vs recreational time, and understand device preferences.

## 2. Objectives

The major objectives of this project are:

- To understand the screentime patterns of users across demographics
- To visualize screen usage across devices, locations, and age groups
- To analyze health impacts due to excessive screen exposure
- To examine the educational-to-recreational usage ratio
- To prepare an interactive dashboard for insights
- To document findings in a final report and presentation

## 3. Dataset Description

Dataset Name: Indian Kids Screentime

2025 Source: Kaggle

Rows:9712

Columns: 8

## Column Details

| Columns                           | Description                                |
|-----------------------------------|--------------------------------------------|
| Age                               | Child's age                                |
| Gender                            | Male/Female                                |
| Avg_Daily_Screen_Time_hr          | Daily hours spent on screens               |
| Primary_Device                    | Mobile, Laptop, Tablet, or TV              |
| Exceeded_Recommended_Limit        | Yes/No                                     |
| Educational_to_Recreational_Ratio | Ratio of productive to entertainment usage |
| Health_Impacts                    | Headache, eye strain, etc.                 |
| Urban_or_Rural                    | Location type                              |

## 4. Data Preprocessing

The following steps were performed based on Week1–Week2 work:

### 4.1 Handling Missing Values

- 3218 missing health impact entries replaced with **“None”**.
- Records with Avg\_Daily\_Screen\_Time\_hr = 0.0 were replaced with **gender-wise mean** values, ensuring realistic representation.

### 4.2 Removing Duplicates

- Duplicate records were identified and removed to avoid bias.

### 4.3 Creating Derived Columns

#### A. Age Bands

Created using `pd.cut()`

Groups: **5–10, 11–15, 16–20**

#### B. Date and Day Type

- Random dates assigned within October 2025

- Day\_Type column derived → **Weekday / Weekend**

### C. Device Share

- Total screen time grouped by Primary\_Device
- Calculated the percentage contribution of each device

### D. Educational & Recreational Time

- Educational Time = Avg\_Screen\_Time × Ratio
- Recreational Time = Total – Educational Time
- Share (%) calculated for visualization

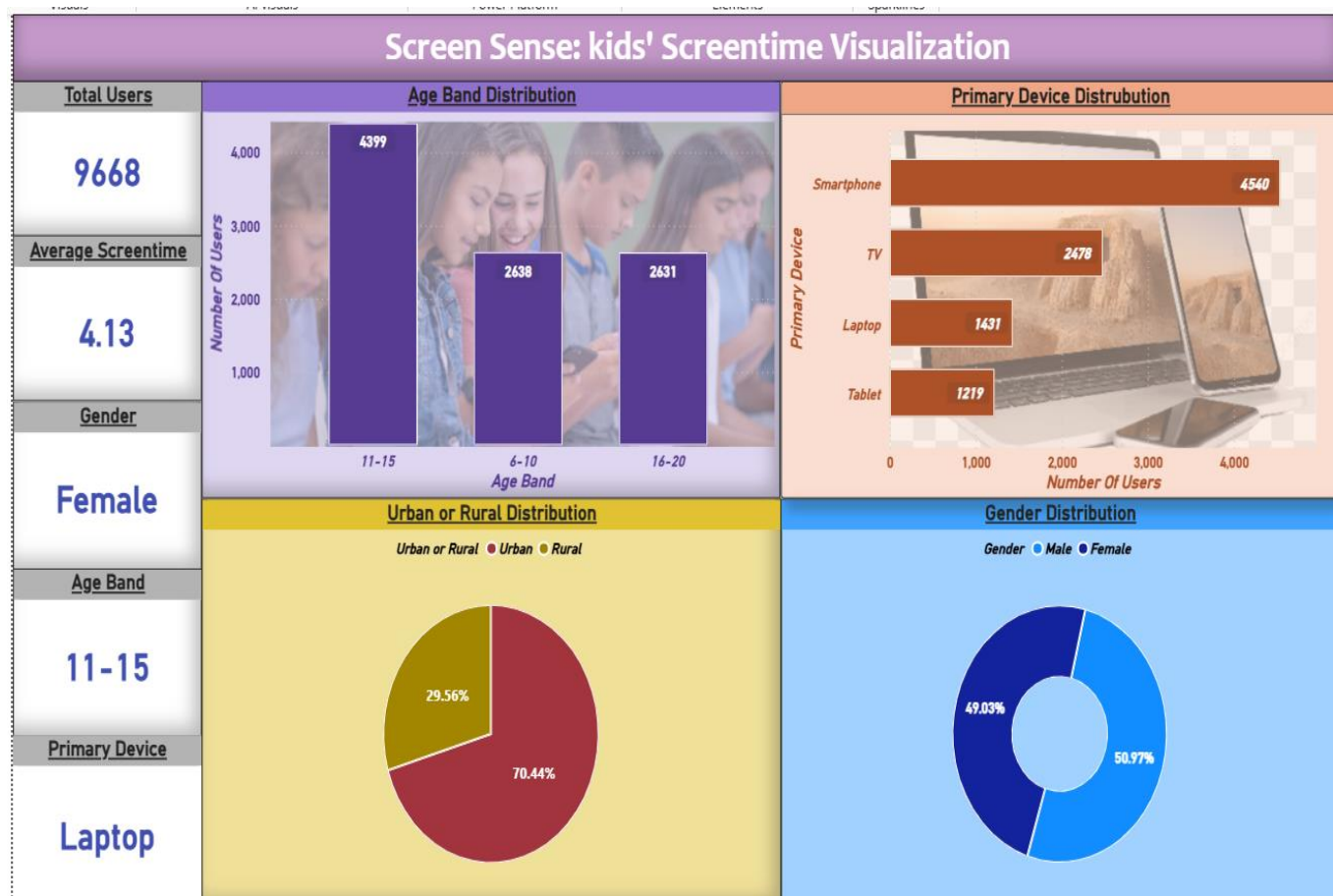
## 4.4 Saving Preprocessed Data

Final cleaned dataset saved as **preprocessed\_data.csv**.

## 5. Dashboard Development

A **two-page Power BI dashboard** was created as the final deliverable.

### Page 1 – Overview Dashboard



## KPI Cards

- 1. Total Users:** This card shows the **total number of participants (9,668)** included in the dataset. It represents the complete sample size used for the screentime analysis.
- 2. Average Screentime:** This KPI displays the **overall average daily screen time (4.13 hours)** across all users. It provides a quick summary of typical digital device usage.
- 3. Gender (Selected Filter Output):** Shows the **currently filtered gender category** from the slicer. In the screenshot, the selected gender is **Female**, meaning all visuals update based on female users.
- 4. Age Band (Selected Filter Output):** Displays the **active age segment** chosen using the slicer. Here, the selected group is **11–15 years**, so all visuals reflect data for that category.
- 5. Primary Device (Selected Filter Output):** Shows the **primary device selected** in the slicer—in this case, **Laptop**. All charts adjust based on this filter.

## Main Dashboard Visuals

### **1. Age Band Distribution (BarChart)**

This bar chart shows the **distribution of users across age groups** (6–10, 11–15, 16–20).

It helps identify **which age group is most active** and how screen time varies between groups.

### **2. Primary Device Distribution (Horizontal Bar Chart)**

This chart displays the **count of users for each primary device** (Smartphone, TV, Laptop, Tablet).

It reveals which device category is used the most, with smartphones being the dominant device.

### **3. Urban or Rural Distribution (Pie Chart)**

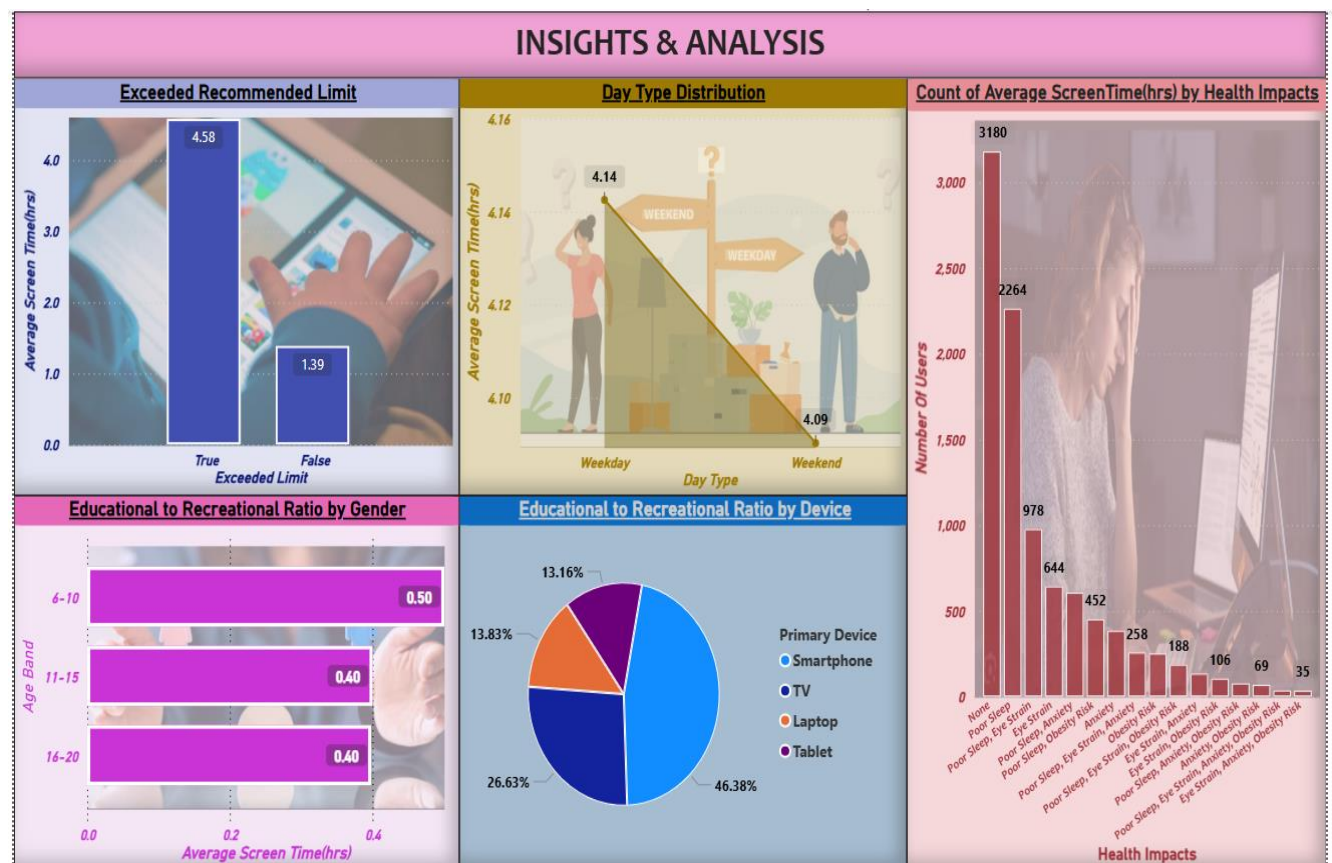
Shows the **percentage split of users based on geographic location**—Urban or Rural.

It helps identify whether screentime trends are more common in urban regions or rural areas.

#### 4. Gender Distribution (Donut Chart)

This donut chart shows the **overall male vs female ratio** in the dataset. It highlights that the dataset is nearly evenly split (approx. 50% each), indicating a balanced gender representation.

### Page 2 – Deep Insights Dashboard



### 1. Exceeded Recommended Limit

- **Definition:** This chart shows the average screen time for users who exceeded the recommended screen time limit versus those who did not.
- **Insight:** Users who exceeded the limit averaged 4.58 hours of screen time, while those within limits averaged 1.39 hours.
- **Purpose:** Helps identify the proportion of users at risk of overexposure to digital devices.

### 2. Day Type Distribution

- **Definition:** Compares average screen time on **weekdays** vs. **weekends**.

- **Insight:** Screen time slightly decreases on weekends (4.09 hrs) compared to weekdays (4.14 hrs), suggesting weekday usage is marginally higher.
- **Purpose:** Shows how daily routines affect digital device usage.

### 3. Count of Average Screen Time (hrs) by Health Impacts

- **Definition:** Bar chart linking average screen time to associated health risks like poor sleep, eye strain, anxiety, and obesity.
- **Insight:** Most users (3180) reported no health impact, while fewer users experienced multiple or severe health risks (e.g., only 35 in highest risk category).
- **Purpose:** Helps assess how screen time correlates with physical and mental health.

### 4. Educational to Recreational Ratio by Gender

- **Definition:** Compares average hours spent on **educational activities** versus **recreational activities** by different age bands.
- **Insight:** Younger age group (6–10) has a slightly higher educational-to-recreational ratio (0.50) than older groups (11–20, 0.40).
- **Purpose:** Helps understand learning versus leisure balance across ages.

### 5. Educational to Recreational Ratio by Device

- **Definition:** Pie chart showing which devices are used primarily for educational versus recreational purposes.
- **Insight:** Smartphones dominate usage (46.38%), followed by laptops (26.63%), TVs (13.83%), and tablets (13.16%).
- **Purpose:** Identifies which devices contribute most to educational or recreational screen time.

## 6. Insights & Observations

### 1. Age-based Usage

- Teenagers (11–15) show the **highest daily screen hours**.
- Younger children (5–10) have lower screen usage.

## 2. Gender-based Usage

- Screen usage is slightly higher among boys across all age groups.

## 3. Device Trends

- Smartphones contribute the **highest share** of total screen time.
- Tablets and laptops have moderate usage.

## 4. Location Patterns

- Urban children have significantly **higher screen exposure** than rural children.

## 5. Educational vs Recreational Time

- Recreational screen time dominates (gaming, entertainment).
- Educational usage decreases with increasing age.

## 6. Health Impacts

- Children crossing 4+ hours/day show more **eye strain, headaches, and fatigue**.

## 7. Weekday vs Weekend

- Weekend screen usage is **25–40% higher** than weekday usage.

## 7. Key Findings

- Most children **exceed recommended limits**, especially mobile users.
- Screen time increases steadily with age
- Recreational activities dominate over educational ones.
- Digital device usage strongly correlates with health complaints.
- Urban lifestyle increases screen usage intensity.

## 8. Conclusion

The ScreenSense project provides a comprehensive view of screentime behavior among children, highlighting crucial patterns in device usage, health impacts, educational engagement, and exceeding recommended screen time.

This dashboard can support parents, educators, and policymakers in making informed decisions and encouraging healthier digital habits.

Future improvements may include integrating machine learning for prediction, age-specific recommendations, and real-time monitoring dashboards.